🥛 Microbe Kitchen

Name: Class: Date:

Activity 1 · Hands-on · 🏠 At home · ⏱ 8–12 hours

🥣 Curd Science

You will learn: Lactic acid bacteria (Lactobacillus) turn milk into curd faster at warm temperature.

YOU NEED

- Milk (3 small cups)
- A spoonful of curd as starter
- Three bowls
- A fridge and a warm place
- Phone microscope

⚠️ **Safety:** Use clean bowls. Do not eat curd that smells bad.

STEPS

1. Warm the milk until lukewarm (not hot).
2. Add half a spoon of curd to each bowl of milk and stir.
3. Keep one bowl in the fridge, one at room temperature, one in a warm place (near the kitchen).
4. Check every 2 hours: is it set? How sour does it smell?
5. Put a drop of curd water on a slide and look at high zoom for tiny rod-shaped bacteria.

OBSERVATION TABLE

Condition	Time to set (h)	Texture	Sourness

QUESTIONS

1. Which bowl set first? Why?

2. What acid makes curd sour?

3. Why is a little curd added to milk?

WHAT IS HAPPENING? — THE SCIENCE

Lactobacillus converts milk sugar (lactose) into lactic acid. The acid makes milk proteins coagulate into curd. Bacteria grow fastest at warm temperatures, so curd sets quickest in a warm place. Lactobacillus also adds vitamin B_{12} and protects the gut.

Key terms: Lactobacillus · lactic acid · fermentation · starter

Activity 2 · Hands-on · 🏠 At home · ⏱ 45 minutes

🎈 Yeast Balloon Race

> **You will learn:** Yeast ferments sugar and releases carbon dioxide — the gas that makes dough rise.

YOU NEED

- 3 small plastic bottles
- 3 balloons
- Dry yeast (2 spoons)
- Sugar
- Warm and cold water

> ⚠️ **Safety:** Do not use hot water — it kills yeast.

STEPS

1. Bottle A: warm water + yeast + sugar. Bottle B: warm water + yeast (no sugar). Bottle C: cold water + yeast + sugar.
2. Shake gently and stretch a balloon over each mouth.
3. Every 10 minutes, measure the balloon's circumference with a string.
4. After 40 minutes compare the three balloons.

OBSERVATION TABLE

Time (min)	Bottle A	Bottle B	Bottle C

QUESTIONS

1. Which balloon grew the most? Why?

2. What gas filled the balloons?

3. Name the yeast used by bakers.

WHAT IS HAPPENING? — THE SCIENCE

Saccharomyces cerevisiae ferments sugar into ethanol and CO_2. Warmth and food (sugar) speed it up. The CO_2 inflates the balloon — the same gas makes bread and idli batter rise; the same process makes alcoholic drinks in industry.

Key terms: **Saccharomyces cerevisiae** **fermentation** **carbon dioxide**

Digital activity

💻 Microbe Matchmaker

> Tap a microbe, then tap what it gives us.

Do this digital activity in the BAIO app: **https://narendra-bio-sir.github.io/BAIO-Learning-System-By-Narendra-Sir/activities.html#pu2-ch8/2**

FOOD · MEDICINE · INDUSTRY CHEMICALS · ENVIRONMENT / FARMING

Food	Medicine	Industry chemicals	Environment / farming

Lactobacillus — curd · Saccharomyces — bread · Propionibacterium — Swiss cheese · Penicillium notatum — penicillin · Trichoderma — cyclosporin A · Monascus — statins · Streptococcus — streptokinase · Aspergillus niger — citric acid · Acetobacter aceti — acetic acid · Clostridium — butyric acid · Methanobacterium — biogas · Rhizobium — nitrogen fixation · Bacillus thuringiensis — biocontrol

> **You will learn:** Microbes are used for food, medicines, industrial chemicals, sewage treatment, biogas, biocontrol and biofertilisers.

THINK & WRITE

1. Which microbe gives a clot-dissolving drug?

2. Name two biofertilisers.